SIH26013 — Simple Teammate Guide

Intelligent Integration and Harmonization of Multi-Source Geospatial Data with Feature-Extracted Cadastral Data

Goal: Understand the problem, the GIS terms, the proposed solution, AI's role, and a realistic 10-hour prototype.

1. The Problem — In Simple Words

Government land information comes from many sources. For the same piece of land, one department may have a cadastral map, another may have municipal records, a survey team may have GNSS points, and a drone survey may provide aerial imagery. These datasets can have different formats, coordinate systems, IDs, boundaries and levels of detail.

Real problem: GIS teams must manually clean, align, compare and connect these datasets. This takes time and can introduce errors.

Think: **many separate datasets about the same land → one consistent geographic view.**

2. Simple Example

Source	Typical information
Cadastral / land records	Parcel boundary, parcel ID, land area
Municipal records	Property and building information
Drone imagery	Actual buildings, roads, walls and visible features
GNSS survey	Accurate geographic survey points
Utility database	Electricity, water or sewer information

The system should determine which records/features refer to the same physical property and bring useful information together.

3. Important Terms

GIS

Geographic Information System — software for storing, analyzing and displaying location-based information on maps.

Geospatial data

Data connected to a geographic location, such as coordinates, roads, buildings or land parcels.

Cadastral data

Digital information about land parcels: boundaries, parcel IDs, area and related land-record information.

Parcel

A specific piece of land represented on a map, usually as a polygon.

Cadastral map

A map showing the boundaries of land parcels.

GeoAI

Using AI/ML together with geographic or remote-sensing data.

Feature extraction

Finding useful physical features from data, such as buildings or roads in aerial imagery.

Orthorectified Imagery (ORI)

Aerial/satellite imagery geometrically corrected so locations line up more accurately with a map.

DSM

Digital Surface Model — elevation of surfaces including buildings, trees and other objects.

DTM

Digital Terrain Model — representation of the underlying terrain/ground elevation.

GNSS

Global Navigation Satellite System. GPS is one example of GNSS.

Ground Truthing (GT)

Checking what actually exists at a location through field/survey verification.

Spatial matching

Using geographic position or geometry to determine which features correspond to each other.

CRS

Coordinate Reference System — the coordinate system used to describe positions on Earth.

GeoJSON

A common format for geographic points, lines and polygons.

GeoTIFF

A geospatial image format that stores imagery together with geographic information.

4. What Solution Does SIH26013 Want?

An intelligent platform that accepts multiple geospatial and non-spatial datasets, standardizes them, extracts useful features, matches related information, and creates a synchronized/combined cadastral view.

Simple flow:

1	**Collect**	Bring in cadastral, imagery, municipal and other datasets.
2	**Standardize**	Align formats and coordinate systems.
3	**Extract**	Use AI/GeoAI to identify useful features from imagery where needed.
4	**Match**	Connect features to the correct parcels using spatial relationships.
5	**Compare**	Find missing information, mismatches and possible conflicts.
6	**Visualize**	Show the result on an interactive GIS map.
7	**Report**	List discrepancies and items requiring officer verification.

5. Where Does AI Fit?

AI is not needed for every part. Traditional GIS is better for coordinate conversion, polygon operations, area calculation and spatial intersections.

Use AI where the computer must understand imagery or messy information.

Input	AI task	Output
Drone/satellite image	Detect buildings	Building locations/polygons
Aerial image	Detect roads/features	Mapped features
Records (optional)	Semantic matching	Likely corresponding records

No external AI API is mandatory. A pretrained open-source computer-vision model can run locally. For a 10-hour prototype, do not train a model from scratch.

6. Example: AI + GIS Together

Suppose the cadastral map contains Parcel P101. The drone image shows a building inside it, but the municipal record says no building is recorded. AI detects the building; GIS checks that it lies inside P101; the system flags the difference.

Result: P101 — possible building-record mismatch — officer verification required.

The prototype should flag possible discrepancies rather than automatically declaring a legal error.

7. What Data Do We Need?

For the prototype, we do **not** need every source mentioned in the PS. A small combination is enough to demonstrate the concept.

Data	Prototype approach	Purpose
Cadastral parcels	Small GeoJSON; synthetic if necessary	Land boundaries
Buildings/roads	OpenStreetMap or extracted features	Reference features
Imagery	Open satellite/drone imagery where available	AI feature extraction
Attributes	Small CSV/GeoJSON table	Non-spatial information

Possible public sources to investigate include OpenStreetMap, Bhuvan, data.gov.in, state/municipal open-data portals and open satellite imagery. Availability and licensing must be checked. If real cadastral data is unavailable, use a clearly labelled synthetic dataset for the prototype.

8. What We Should Build in 10 Hours

Priority	Feature	Do it?
1	Parcel GeoJSON + map	YES
2	Reference imagery/features	YES
3	Pretrained AI building detection	YES, if it works quickly
4	Building → parcel spatial matching	YES
5	Simple discrepancy detection	YES
6	Dashboard statistics	YES
7	PDF/report export	Optional
8	GNSS + DSM + DTM + all government integrations	NO
9	Custom AI model training	NO

9. Recommended Technology

Dashboard: Streamlit + Folium/Leaflet
Processing: Python
GIS: GeoPandas + Shapely + GDAL
Database: PostgreSQL + PostGIS (optional for MVP)
AI: pretrained YOLO/segmentation model with PyTorch/Ultralytics
Formats: GeoJSON and GeoTIFF

10. Architecture

DATA SOURCES → PREPROCESSING → CRS/FORMAT ALIGNMENT → AI FEATURE EXTRACTION → SPATIAL MATCHING → HARMONIZATION → DISCREPANCY DETECTION → GIS DASHBOARD → REPORT

Key separation: AI extracts/understands features; GIS performs deterministic geographic operations; officers verify important discrepancies.

11. 10-Hour Work Plan

Time	Task
0–1 hr	Set up Python + Streamlit.
1–2 hr	Prepare small parcel/reference dataset.
2–4 hr	Integrate pretrained building/feature model if feasible.
4–5.5 hr	Convert AI output to geographic features and match to parcels.
5.5–7 hr	Add discrepancy rules and statistics.
7–9 hr	Build interactive GIS dashboard.
9–10 hr	Test, polish demo and prepare presentation.

12. Demo Flow

1. Show three datasets coming from different sources.

2. Load the parcel layer.

3. Display imagery/reference features.

4. Run AI feature extraction.

5. Show detected buildings/features.

6. Automatically associate features with parcels.

7. Show one or two deliberate discrepancies.

8. Display them on the map and in a table.

9. Explain that the officer gets one harmonized view instead of manually comparing separate datasets.

13. What NOT to Claim

Do not claim that a 10-hour prototype fully integrates India's land records or replaces government GIS workflows. Present it as a proof-of-concept for automated multi-source harmonization, feature extraction, spatial matching and discrepancy detection.

14. One-Sentence Explanation for Teammates

“Different government departments have different data about the same land. We are building an AI + GIS system that brings those datasets onto a common map, automatically identifies physical features, connects them to the correct parcels, and flags inconsistencies for officers to verify.”

15. Final Understanding

The problem is not simply 'make a map.' The important part is **automatically making different land-related datasets work together**. AI handles difficult visual/semantic tasks, GIS handles geographic calculations, and the final system gives officers a unified, traceable view.